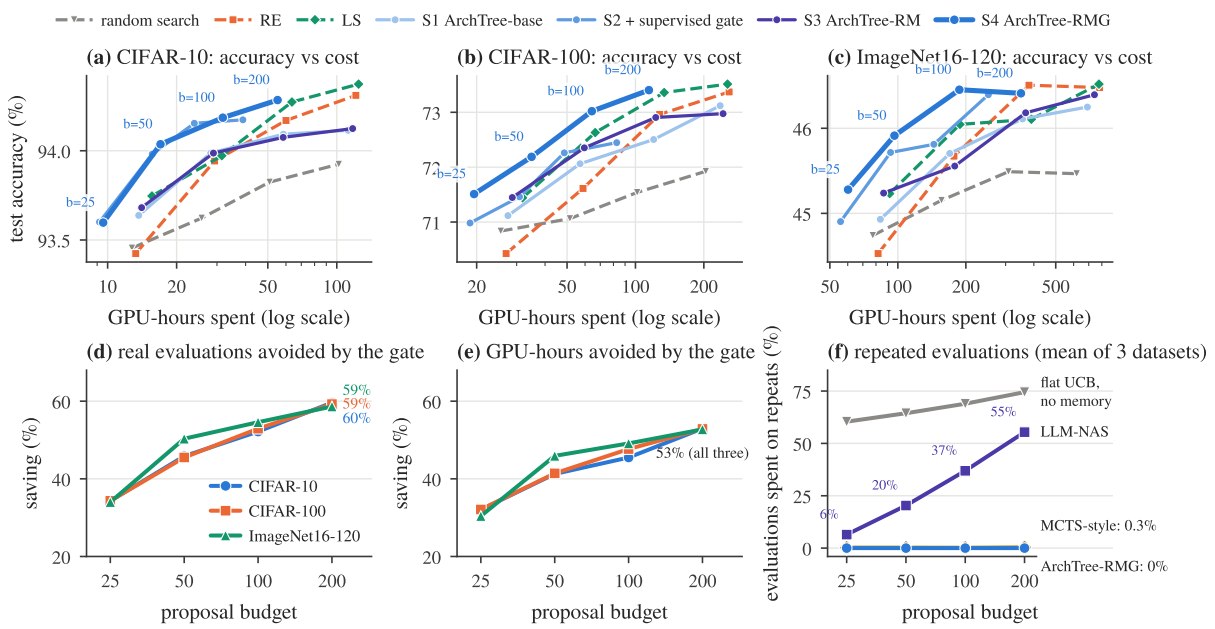
random search
RE
LS
S1 ArchTree-base
S2 + supervised gate
S3 ArchTree-RM
S4 ArchTree-RMG
(a) CIFAR-10: accuracy vs cost
(b) CIFAR-100: accuracy vs cost
(c) ImageNet16-120: accuracy vs cost
test accuracy (%)
GPU-hours spent (log scale)
b=25
b=50
b=100
b=200
(d) real evaluations avoided by the gate
(e) GPU-hours avoided by the gate
(f) repeated evaluations (mean of 3 datasets)
saving (%)
proposal budget
CIFAR-10
CIFAR-100
ImageNet16-120
59%
59%
60%
53% (all three)
evaluations spent on repeats (%)
flat UCB, no memory
LLM-NAS
MCTS-style: 0.3%
ArchTree-RMG: 0%
55%
37%
20%
6%